

ASHUTOSH ADHIKARI

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PROFILE

Backend-focused Machine Learning Engineer with extensive experience in designing, building, and deploying production-grade ML systems and APIs. Specialized in FastAPI-based backend services, Dockerized microservices, and scalable cloud deployments on AWS. Proficient in NLP, LLM-powered search systems, and enterprise data platforms with strong emphasis on reliability, performance, and maintainability. Currently developing agentic execution platforms with browser automation and workflow orchestration.

WORK EXPERIENCE

TAI Inc. **January 2022 - Present**
Backend/Machine Learning Engineer

- Designed and developed backend ML services using FastAPI to expose search, NLP, and inference APIs for enterprise applications.
- Built and maintained Dockerized microservices for ML models and backend workflows with production deployment.
- Developed ML-powered search systems for Nomura Research Institute (Japan's top fintech company).
- Designed scalable data ingestion, indexing, and retrieval pipelines for large Japanese text datasets.
- Implemented model training and inference pipelines using AWS SageMaker with CI/CD automation.
- Integrated AWS services (EC2, S3, ECR, IAM, CloudWatch) for secure and observable deployments.
- Researched and integrated LLM-based solutions for semantic search and NLP use cases.
- Applied Explainable AI (XAI) techniques to improve transparency of transformer-based models.
- Collaborated with frontend teams to support graph-based visualization using React and Sigma.js.
- Currently developing agentic execution platform with Playwright automation and FastAPI orchestration.

TAI Inc. **September 2021 - December 2021**
Machine Learning Intern

- Conducted research on integrating machine learning techniques with music-based applications.
- Implemented and evaluated state-of-the-art AI techniques for music analysis and synthesis.

CloudFactory **July 2020 - August 2021**
Annotation QA

- Ensured quality and consistency of annotated image datasets for machine learning models.
- Reviewed and validated labeling accuracy against project standards.

TECHNICAL SKILLS

Backend & APIs	FastAPI, Flask, REST APIs, Async Python
Programming Languages	Python, JavaScript, SQL, Java
Machine Learning & NLP	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LLMs, LangChain, FAISS, XAI
Cloud & DevOps	AWS (SageMaker, EC2, S3, ECR, IAM, CloudWatch), Docker
Databases & Search	Neo4j, Elasticsearch, MongoDB, PostgreSQL
Frontend	React, Sigma.js, SurveyJS
CI/CD Pipelines	Jenkins, GitLab CI/CD
Tools & Platforms	Git, Jupyter, Pandas, Matplotlib, Seaborn, Playwright, VNC

EDUCATION

The British College

2017 - 2021

Bachelors of Science in Information Technology (BSc IT)

Relevant Courses: Data Structures, Discrete Systems, Cognitive Science, Artificial Intelligence, Database Administration

PROJECTS

Agentic QA

Python, FastAPI, Docker, Playwright, AWS

- Developing backend-driven platform to orchestrate automated browser-based workflows.
 - Designing FastAPI services for job scheduling, execution lifecycle, and artifact management.
 - Integrating Playwright for browser automation with headless/headed execution support.
 - Implementing Docker-based isolated execution environments for reproducibility and security.
 - Enabling GUI-based execution via VNC for real-time visualization and debugging.
 - Architecting system to support AI-driven agents and automated testing workflows.

TabSense (AI Browser Assistant)

Chrome Web Store

React, TensorFlow.js, Chrome Manifest V3, RAG

- Developed and published a privacy-focused Chrome extension for AI-powered tab management and research.
 - Architected a Manifest V3 extension using React and Webpack with a non-blocking service worker architecture.
 - Implemented on-device machine learning using TensorFlow.js (Universal Sentence Encoder) to classify tabs.
 - Designed an adaptive online learning system that incrementally retrains the local model based on user manual grouping actions.
 - Integrated streaming LLM APIs (OpenAI, OpenRouter) with real-time token streaming and Markdown rendering.

Visa Processing System

Python, FastAPI, SurveyJS, MongoDB, AWS

- Developed complete visa application processing system with dynamic form handling.
 - Integrated SurveyJS for dynamic, customizable visa application forms with validation.
 - Built RESTful APIs using FastAPI for collecting and processing application data.
 - Implemented MongoDB for storing form responses and application data.
 - Created admin dashboard for processing applications and generating reports.
 - Designed secure file upload system with AWS S3 integration.

Aslead Search

Python, FastAPI, PyTorch, AWS SageMaker

- ML-powered search application for Nomura Research Institute.
 - Built technical document classification system achieving approximately 95% accuracy.
 - Implemented transformer-based models for document classification and semantic similarity.
 - Integrated LLM-based approaches for enterprise search use cases.
 - Applied XAI techniques for model interpretability.
 - Developed scalable training and inference pipelines on AWS SageMaker.

Relative Analyzer

React, Sigma.js, Neo4j, Docker

- Web application for visualizing graph data stored in Neo4j.
 - Developed visualization interface using React and Sigma.js.
 - Optimized Neo4j queries, improving performance by over 50×.
 - Executed Python services inside Docker containers on client servers.
 - Integrated Elasticsearch for enhanced search capabilities.

Melody Synthesis

Python, Magenta.js, TensorFlow

- Prototype system for music synthesis using machine learning.
 - Built ML-based system to synthesize new melodies.
 - Evaluated and implemented state-of-the-art AI research for music generation.
 - Developed user-friendly interface for music synthesis.